

The World Before Brains

LIFE EXISTED ON Earth for a long time—and I mean a *long* time, over three billion years—before the first brain made an appearance. By the time the first brains evolved, life had already persevered through countless evolutionary cycles of challenge and change. In the grand arc of life on Earth, the story of brains would not be found in the main chapters but in the epilogue—brains appeared only in the most recent 15 percent of life’s story. Intelligence too existed for a long time before brains; as we will see, life began exhibiting intelligent behavior early in its story. We cannot understand why and how brains evolved without first reviewing the evolution of intelligence itself.

Around four billion years ago, deep in the volcanic oceans of a lifeless Earth, just the right soup of molecules were bouncing around the microscopic nooks and crannies of an unremarkable hydrothermal vent. As boiling water burst from the seafloor, it smashed naturally occurring nucleotides together and transformed them into long molecular chains that closely resembled today’s DNA. These early DNA-like molecules were short-lived; the same volcanic kinetic energy that constructed them also inevitably ripped them apart. Such is the consequence of the second law of thermodynamics. That unbreakable law of physics which declares that entropy—the amount of disorder in a system—always and unavoidably increases; the universe cannot help but tend toward decay. After countless random nucleotide chains were constructed and destroyed, a lucky sequence was stumbled upon, one that marked, at least on Earth, the first true rebellion against the seemingly inexorable onslaught of entropy. This new DNA-like molecule wasn’t alive per se, but it performed the most fundamental process by which life would later emerge: it duplicated itself.

Although these self-replicating DNA-like molecules also succumbed to the destructive effects of entropy, they didn’t have to survive *individually* to

survive *collectively*—as long as they endured long enough to create their own copies, they would, in essence, persist. This is the genius of self-replication. With these first self-replicating molecules, a primitive version of the process of evolution began; any new lucky circumstances that facilitated more successful duplication would, of course, lead to more duplicates.

There were two subsequent evolutionary transformations that led to life. The first was when protective lipid bubbles entrapped these DNA molecules using the same mechanism by which soap, also made of lipids, naturally bubbles when you wash your hands. These DNA-filled microscopic lipid bubbles were the first versions of cells, the fundamental unit of life.

The second evolutionary transformation occurred when a suite of nucleotide-based molecules—ribosomes—began translating specific sequences of DNA into specific sequences of amino acids that were then folded into specific three-dimensional structures we call *proteins*. Once produced, these proteins float around inside a cell or are embedded in the wall of the cell fulfilling different functions. You have probably, at least in passing, heard that your DNA is made up of genes. Well, a gene is simply the section of DNA that codes for the construction of a specific and singular protein. This was the invention of protein synthesis, and it is here that the first sparks of intelligence made their appearance.

DNA is relatively inert, effective for self-duplication but otherwise limited in its ability to manipulate the microscopic world around it. Proteins, however, are far more flexible and powerful. In many ways, proteins are more machine than molecule. Proteins can be constructed and folded into many shapes—sporting tunnels, latches, and other robotic moving parts—and can thereby subserve endless cellular functions, including “intelligence.”

Even the simplest single-celled organisms—such as bacteria—have proteins designed for movement, motorized engines that convert cellular energy into propulsion, rotating propellers using a mechanism no less complex than the motor of a modern boat. Bacteria also have proteins designed for *perception*—receptors that reshape when they detect certain features of the external environment, such as temperature, light, or touch.

Armed with proteins for movement and perception, early life could monitor and respond to the outside world. Bacteria can swim away from environments that lower the probability of successful replication,

environments that have, for example, temperatures that are too hot or cold or chemicals that are destructive to DNA or cell membranes. Bacteria can also swim toward environments that are amenable to reproduction.

And in this way, these ancient cells indeed had a primitive version of intelligence, implemented not in neurons but in a complex network of chemical cascades and proteins.

The development of protein synthesis not only begot the seeds of intelligence but also transformed DNA from mere *matter* to a medium for storing *information*. Instead of being the self-replicating stuff of life itself, DNA was transformed into the informational foundation from which the stuff of life is constructed. DNA had officially become life's blueprint, ribosomes its factory, and proteins its product.

With these foundations in place, the process of evolution was initiated in full force: variations in DNA led to variations in proteins, which led to the evolutionary exploration of new cellular machinery, which, through natural selection, were pruned and selected for based on whether they further supported survival. By this point in life's story, we have concluded the long, yet-to-be-replicated, and mysterious process scientists call abiogenesis: the process by which nonbiological matter (*abio*) is converted into life (*genesis*).

The Terraforming of Earth

Shortly thereafter, these cells evolved into what scientists call the “last universal common ancestor,” or LUCA. LUCA was the genderless grandparent of all life; every fungus, plant, bacteria, and animal alive today, including us, descend from LUCA. It is no surprise, then, that all life shares the core features of LUCA: DNA, protein synthesis, lipids, and carbohydrates.

LUCA, living around 3.5 billion years ago, likely resembled a simpler version of a modern bacteria. And indeed, for a long time after this, all life was bacterial. After a further billion years—through trillions upon trillions of evolutionary iterations—Earth's oceans were brimming with many diverse species of these microbes, each with its own portfolio of DNA and proteins. One way in which these early microbes differed from one another was in their systems of energy production. The story of life, at its core, is as much about energy as it is about entropy.

Keeping a cell alive is expensive. DNA requires constant repair; proteins require perpetual replenishment; and cellular duplication requires a reconstruction of many inner structures. Hydrogen, an element abundant near hydrothermal vents, was likely the first fuel used to finance these many processes. But this hydrogen-based energy system was inefficient and left life desperately grasping for enough energy to survive. After more than a billion years of life, this energetic poverty came to an end when a single species of bacteria—the cyanobacteria, also called blue-green algae—found a far more profitable mechanism for extracting and storing energy: photosynthesis.

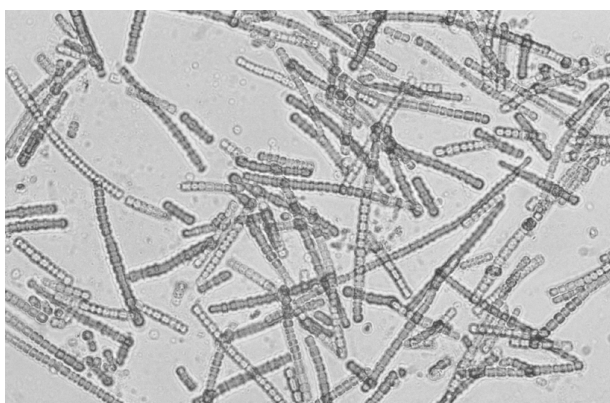


Figure 1.1: Cyanobacteria

Photograph by Willem van Aken on March 18, 1993. Figure from www.scienceimage.csiro.au/image/4203 CC BY 3.0 license.

The most impressive biological system in these early cyanobacteria was not their protein factories or their products but their photosynthetic power plants—the structures that converted sunlight and carbon dioxide into sugar, which could then be stored and converted into cellular energy. Photosynthesis was more efficient than prior cellular systems for extracting and storing energy. It provided cyanobacteria with an abundance of fuel with which to finance their duplication. Vast regions of the ocean quickly became covered in greenish gooey microbial mats—colonies of billions of cyanobacteria basking in the sun, sucking up carbon dioxide, and endlessly reproducing.

Like most processes of energy production, from burning fossil fuels to harnessing nuclear fuels, photosynthesis produced a pollutive exhaust. Instead of carbon dioxide or nuclear waste, the exhaust from photosynthesis was oxygen. Before this time, the Earth had no ozone layer. It was the cyanobacteria, with their newfound photosynthesis, that constructed Earth's oxygen-rich atmosphere and began to terraform the planet from a gray volcanic rock to the oasis we know today. This event happened around 2.4 billion years ago, and it occurred so rapidly, at least in geological terms, that it has been called the Great Oxygenation Event. Over the course of one hundred million years, oxygen levels skyrocketed. Unfortunately, this event was not a boon for all life. Scientists have a less forgiving name for it: The Oxygen Holocaust.

Oxygen is an incredibly reactive element, which makes it dangerous in the carefully orchestrated chemical reactions of a cell. Unless special intracellular protective measures are taken, oxygen compounds will interfere with cellular processes, including the maintenance of DNA. This is why antioxidants—compounds that remove highly reactive oxygen molecules from your bloodstream—are believed to offer protection from cancer. The photosynthetic life-forms became victims of their own success, slowly suffocating in a cloud of their own waste. The rise of oxygen was followed by one of the deadliest extinction events in Earth's history.

As with many substances that are dangerous (uranium, gasoline, coal), oxygen can also be useful. This newly available element presented an energetic opportunity, and it was only a matter of time before life stumbled onto a way to exploit it. A new form of bacteria emerged that produced energy not from photosynthesis but from cellular respiration—the process by which oxygen and sugar is converted into energy, spewing out carbon dioxide as exhaust. Respiring microbes began gobbling up the ocean's excess oxygen and replenishing its depleted supply of carbon dioxide. What began as a pollutant to one form of life became fuel for another.

Life on Earth fell into perhaps the greatest symbiosis ever found between two competing but complementary systems of life, one that lasts to this day. One category of life was photosynthetic, converting water and carbon dioxide into sugar and oxygen. The other was respiratory, converting sugar and oxygen back into carbon dioxide. At the time, these two forms of life were similar, both single-celled bacteria. Today this symbiosis is made up of very different forms of life. Trees, grass, and other plants are some of

our modern photosynthesizers, while fungi and animals are some of our modern respirators.

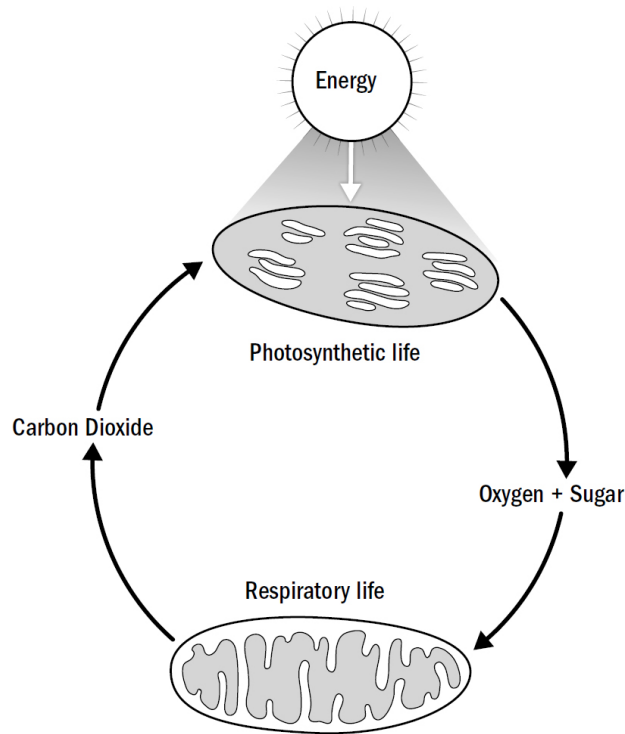


Figure 1.2: The symbiosis between photosynthetic and respiratory life

Original art by Rebecca Gelernter

Cellular respiration requires sugar to produce energy, and this basic need provided the energetic foundation for the eventual intelligence explosion that occurred uniquely within the descendants of respiratory life. While most, if not all, microbes at the time exhibited primitive levels of intelligence, it was only in respiratory life that intelligence was later elaborated and extended. Respiratory microbes differed in one crucial way from their photosynthetic cousins: they needed to hunt. And hunting required a whole new degree of smarts.

Three Levels

The ecosystem two billion years ago was not a particularly war-torn world.*
A tentative peace brokered by energetic necessity undergirded life's many

interactions. Although some bacteria might have gobbled up the remains of nearby deceased neighbors, it was rarely worth the effort for them to actively try to kill other life. The non-oxygen-based approach to converting sugar into energy (*anaerobic* respiration) is fifteen times less efficient than the oxygen-based approach (*aerobic* respiration). As such, before the introduction of oxygen, hunting was not a viable survival strategy. It was better to just find a good spot, sit tight, and bask in the sunlight. The most severe competition between early life was likely akin to people rushing into a Walmart for a Black Friday sale, elbowing others out of the way for scarce nearby prizes but not directly assaulting one another. Even such elbowing was probably uncommon; sunlight and hydrogen were abundant, and there was more than enough to go around.

However, unlike the cells that came before, respiratory life could survive only by stealing the energetic prize—the sugary innards—of photosynthetic life. Thus, the world’s utopic peace ended quite abruptly with the arrival of aerobic respiration. It was here that microbes began to actively eat other microbes. This fueled the engine of evolutionary progress; for every defensive innovation prey evolved to stave off being killed, predators evolved an offensive innovation to overcome that same defense. Life became caught in an arms race, a perpetual feedback loop: offensive innovations led to defensive innovations that required further offensive innovations.

Out of this maelstrom, a large diversification of life emerged. Some species remained small single-celled microbes. Other species evolved into the first eukaryotes (pronounced “you-care-ee-oats”), cells that were over a hundred times larger, produced a thousand times more energy, and had much more internal complexity. These eukaryotes were the most advanced microbial killing machines yet. Eukaryotes were the first to evolve *phagotrophy*—the hunting strategy of literally engulfing other cells and breaking them down inside their cellular walls. These eukaryotes, armed with more energy and complexity, further diversified into the first plants, the first fungi, and the precursors of the first animals. The fungi and animal descendants of eukaryotes retained their need to *hunt* (they were respirators), while plant lineage returned to a lifestyle of photosynthesis.

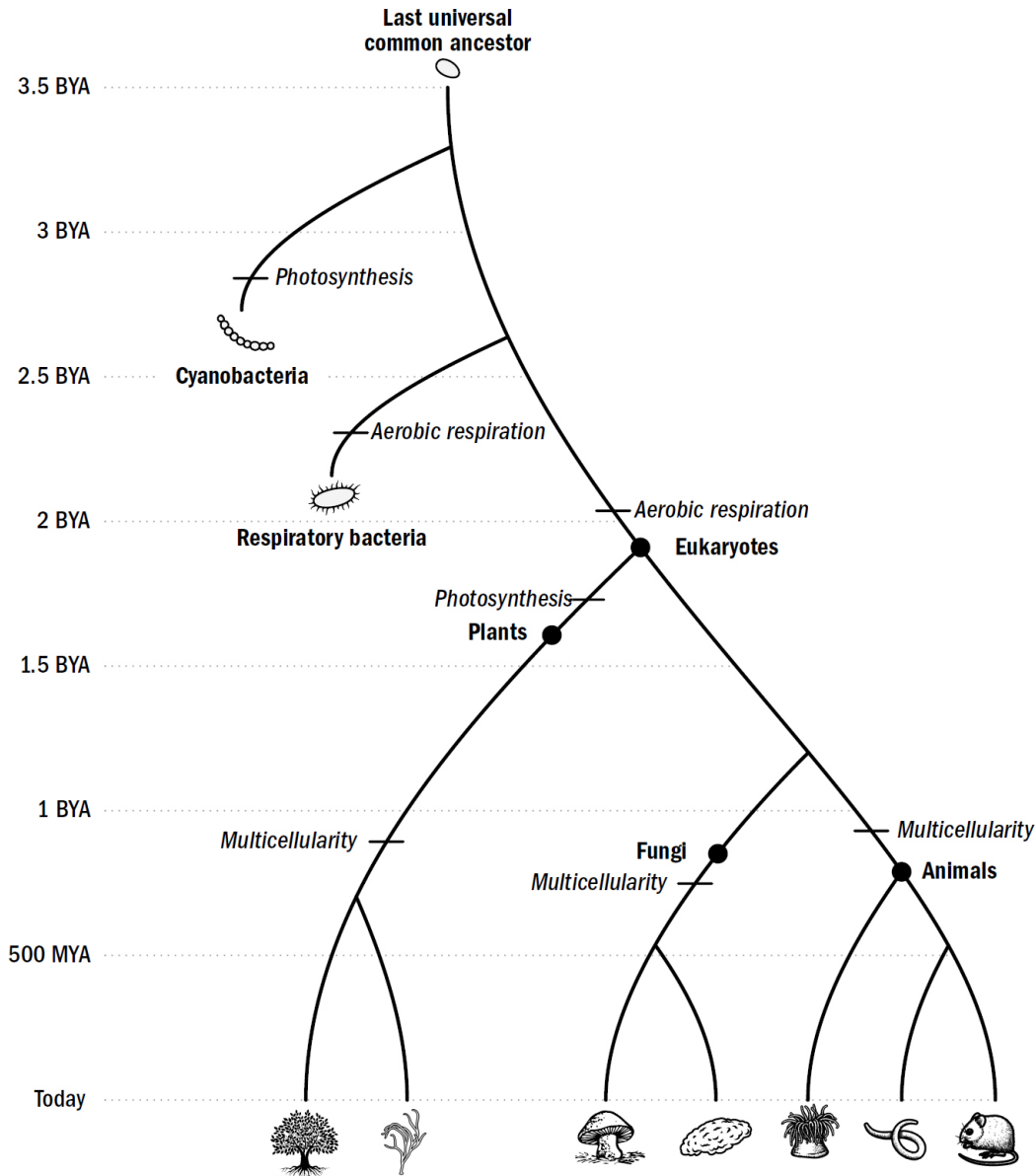


Figure 1.3: The tree of life
Original art by Rebecca Gelernter

What was common across these eukaryote lineages was that all three—plants, fungi, and animals—each independently evolved multicellularity. Most of what you see and think of as life—humans, trees, mushrooms—are primarily *multicellular* organisms, cacophonies of billions of individual cells all working together to create a singular emergent organism. A human is made up of exactly such diverse types of specialized cells: skin cells, muscle cells, liver cells, bone cells, immune cells, blood cells. A plant has

specialized cells too. These cells all serve different *functions* while still serving a common *purpose*: supporting the survival of the overall organism.

And so, seaweed-like underwater plants began sprouting, mushroom-like fungi began growing, and primitive animals began slowly swimming around. By about eight hundred million years ago, life would have fallen into three broad levels of complexity. At level one, there was single-celled life, made up of microscopic bacteria and single-celled eukaryotes. At level two, there was small multicellular life, large enough to engulf single-celled organisms but small enough to move around using basic cellular propellers. At level three, there was *large* multicellular life; too big to move with cellular propellers, and therefore forming immobile structures.

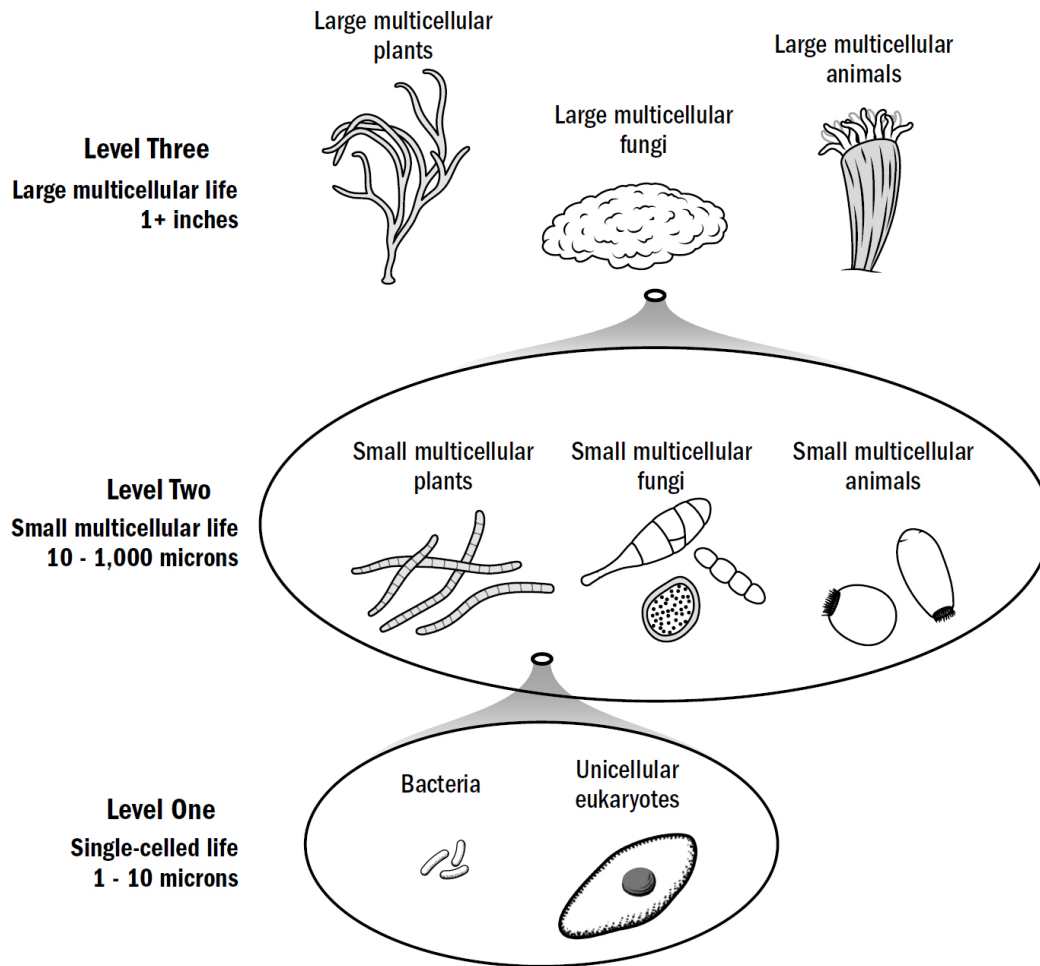


Figure 1.4: Three complexity levels in the ancient sea before brains

Original art by Rebecca Gelernter

These early animals probably wouldn't resemble what you think of as animals. But they contained something that made them different from all other life at the time: neurons.

The Neuron

What a neuron is and what it does depends on whom you ask. If you ask a biologist, neurons are the primary cells that make up the nervous system. If you ask a machine learning researcher, neurons are the fundamental units of neural networks, little accumulators that perform the basic task of computing a weighted summation of their inputs. If you ask a psychophysicist, neurons are the sensors that measure features of the

external world. If you ask a neuroscientist specializing in motor control, neurons are effectors, the controllers of muscles and movement. If you ask other people, you might get a wide range of answers, from “Neurons are little electrical wires in your head” to “Neurons are the stuff of consciousness.” All of these answers are right, carrying a kernel of the whole truth, but incomplete on their own.

The nervous systems of all animals—from worms to wombats—are made up of these stringy odd-shaped cells called neurons. There is an incredible diversity of neurons, but despite this diversity in shapes and sizes, all neurons work the same way. This is the most shocking observation when comparing neurons across species—they are all, for the most part, fundamentally identical. The neurons in the human brain operate the same way as the neurons in a jellyfish. What separates you from an earthworm is not the unit of intelligence itself—neurons—but how these units are wired together.

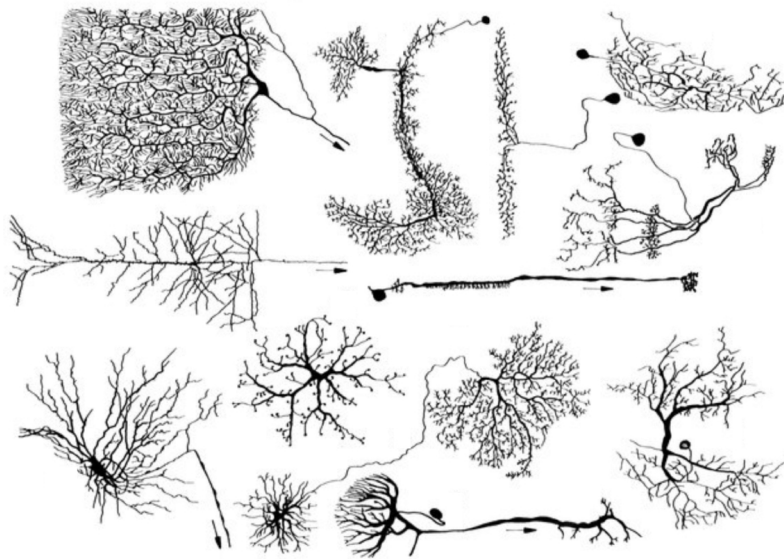


Figure 1.5: Neurons

Figure from Reichert, 1990. Used with permission.

Animals with neurons share a common ancestor, an organism in whom the first neurons evolved and from whom all neurons descend. It seems that in this ancient grandmother of animals, neurons attained their modern form; from this point on, evolution rewired neurons but made no meaningful adjustments to the basic unit itself. This is a glaring example of how prior

innovations impose constraints on future innovations, often leaving early structures unchanged—the fundamental building blocks of brains have been essentially the same for over six hundred million years.

Why Fungi Don't Have Neurons but Animals Do

You aren't all that different from mold. Despite their appearance, fungi have more in common with animals than they do with plants. While plants survive by photosynthesis, animals and fungi both survive by respiration. Animals and fungi both breathe oxygen and eat sugar; both digest their food, breaking cells down using enzymes and absorbing their inner nutrients; and both share a much more recent common ancestor than either do with plants, which diverged much earlier. At the dawn of multicellularity, fungi and animal life would have been extremely similar. And yet one lineage (animals) went on to evolve neurons and brains, and the other (fungi) did not. Why?

Sugar is produced only by life, and thus there are only two ways for large multicellular respiratory organisms to feed. One is to wait for life to die, and the other is to catch and kill living life. Early in the fungi-animal divergence, they each settled into opposing feeding strategies. Fungi chose the strategy of waiting, and animals chose the strategy of killing.* Fungi eat through *external* digestion (secreting enzymes to break food down outside the body), while animals eat through *internal* digestion (trapping food inside the body and then secreting enzymes). The fungal strategy was, by some measures, more successful than the animals'—by biomass, there is about six times more fungus on Earth than animals. But as we will continually see, it is usually the worse strategy, the harder strategy, from which innovation emerges.

Fungi produce trillions of single-celled spores that float around dormant. If by luck one happens to find itself near dying life, it will blossom into a large fungal structure, growing hairy filaments into the decaying tissue, secreting enzymes, and absorbing the released nutrients. This is why mold always shows up in old food. Fungal spores are all around us, patiently waiting for something to die. Fungi are currently, and likely have always been, Earth's garbage collectors.

Early animals, however, settled into a strategy of actively catching and ingesting level-two multicellular prey (see [figure 1.4](#)). Active killing was, of